

Two questions about a house

regression, classification, and how to check yourself

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About me

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My stack

These slides were designed by me and implemented by **DeepSeek V4.1 Flash**.



DeepSeek

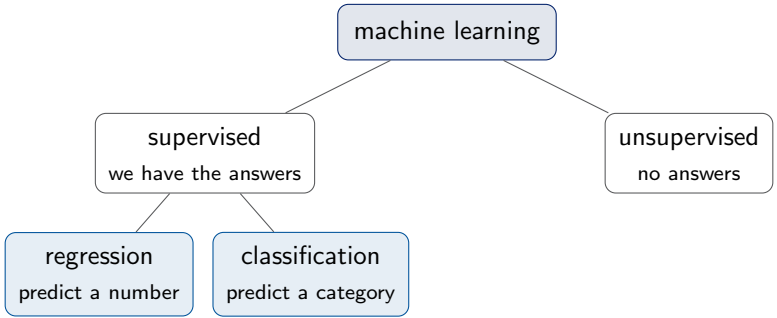


OpenCode

the stack I use for coding

What kind of machine learning?

- regression
- classification



Two questions about the same houses

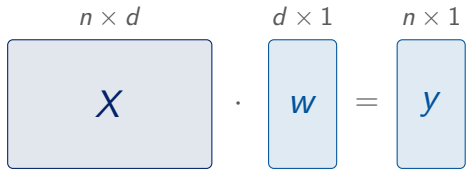
- regression
- classification

size	beds	age	neighborhood	price	sold
1800	3	12	Westside	420	yes
2400	4	5	Northgate	610	yes
1200	2	40	Westside	275	no
3000	5	2	Hillcrest	780	no
1500	3	25	Northgate	350	yes

price $\rightarrow$ regression (a number) $\cdot$ sold $\rightarrow$ classification (a category)

The data is two matrices

- regression
- classification



$$(n \times d)\,(d \times 1) \;=\; (n \times 1)$$

one row = one house · we pick w so Xw lands as close to y as possible

CATALYST

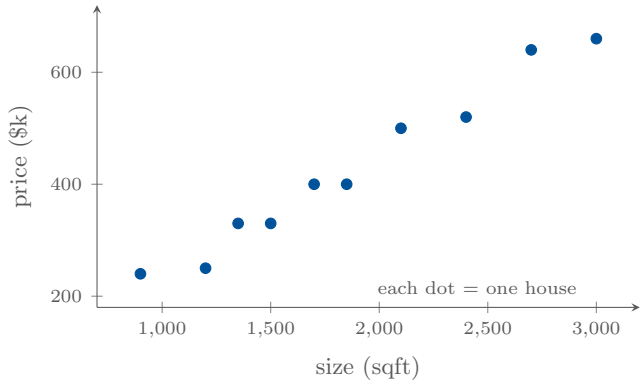
- regression
- classification

1 The models

a line, then a curve

Price from size

- regression
- classification

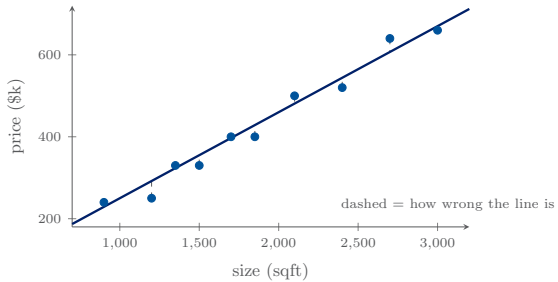


bigger house → higher price?

CATALYST

Draw the line that fits best

- **regression**
- classification



$$\hat{y} = w\,x + b \quad w \text{ tilts it, } b \text{ slides it}$$

CATALYST

More columns, more dials

feature	value	weight
size	1800	w_1
bedrooms	3	w_2
age	12	w_3
add them up + b		$\hat{y}$

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + b = w \cdot x + b$$

the same thing with matrices: $\hat{y} = Xw + b$

How wrong is the line?

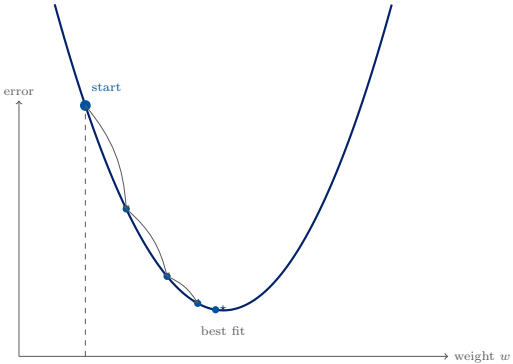
$$\mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} \left(y_i - \hat{y}_i \right)^2$$

y_i	the real price
$\hat{y}_i$	the price the line guesses
$y_i - \hat{y}_i$	the miss

square the misses so they don't cancel, then average

How it learns

- regression
- classification



step downhill a little, repeat:

$$w \leftarrow w - \eta \frac{2}{n} X^\top (Xw - y)$$

η is how big a step

CATALYST

There's also a shortcut

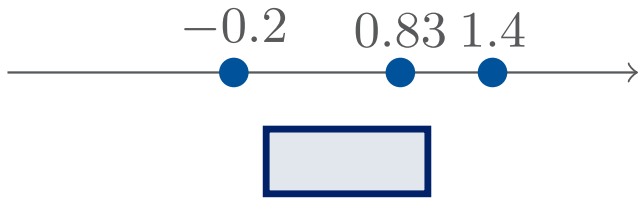
$$w = (X^\top X)^{-1} X^\top y$$

the exact answer in one shot — no stepping needed

one feature: $w = \dfrac{\mathrm{cov}(x, y)}{\mathrm{var}(x)}, \quad b = \bar{y} - w\bar{x}$

But a line gives a number

- regression
- classification

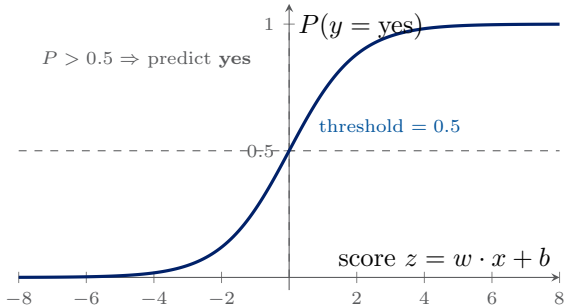


a probability lives in here

for "sold?" we want a yes/no, not a 1.4

Squeeze it into 0 to 1

- regression
- classification

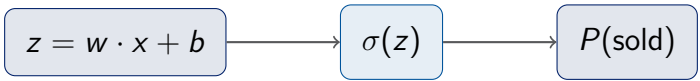


$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

CATALYST

The model

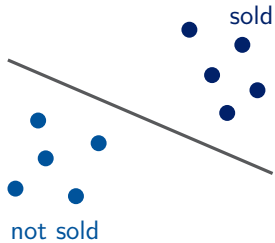
- regression
- classification



$$P(\text{sold}) = \sigma(w \cdot x + b)$$

Where the answer flips

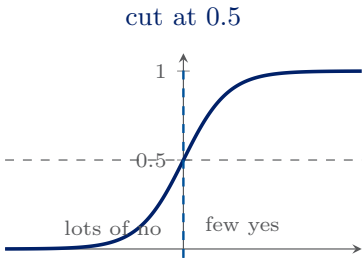
- regression
- classification

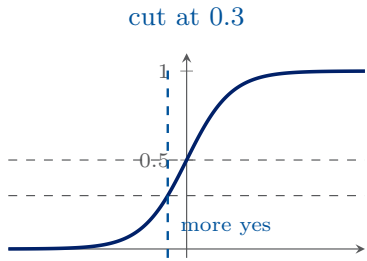


$w \cdot x + b = 0$ where the guess flips from no to yes

CATALYST

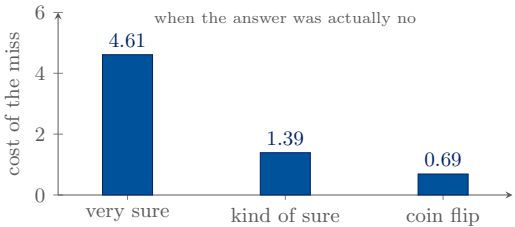
Where you cut is your call





lower the cut → catch more sold, but more false alarms

What a miss costs



$$-\frac{1}{n}\sum\Big[y\log p+(1-y)\log(1-p)\Big]$$

sure and wrong costs a lot; unsure and wrong costs a little

- regression
- classification

2
Features

models only eat numbers

A feature is one column

- regression
- classification

size	bedrooms	neighborhood
1800	3	Westside
2400	4	Northgate

one thing we measured about each house

Models only eat numbers

- regression
- classification



not a number

so we turn text into numbers — in a way that keeps the meaning

One-hot: one text column becomes several 0/1 colu

- regression
- classification

neighborhood
Westside
Northgate
Hillcrest

$\longrightarrow$

West	North	Hill
1	0	0
0	1	0
0	0	1

a 1 marks which neighborhood it is

Ordered text and different scales

- regression
- classification

ordinal — there's a real order:

poor	0
fair	1
good	2

scaling — put them on one scale:

$$z = \frac{x - \mu}{\sigma}$$

sqft and bedrooms shouldn't fight

- regression
- classification

3

Checking your work

splits, folds, and scores

Never test on what you trained on

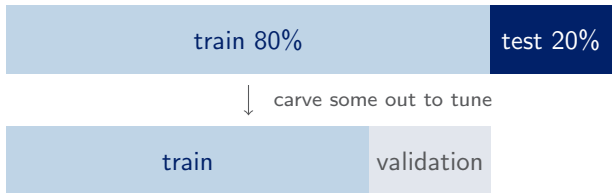
- regression
- classification



learn on the training part · score on the part you never looked at

Split it 80/20

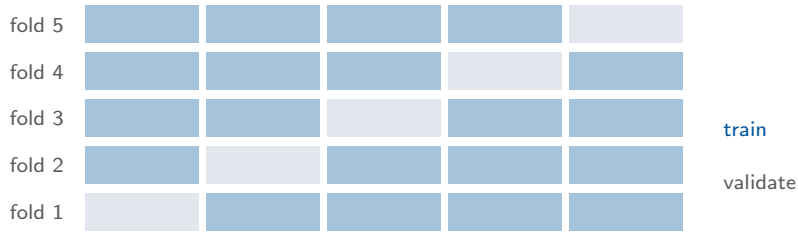
- regression
- classification



we hand you the training rows — split your own test set off first

Five-fold cross-validation

- regression
- classification



train on 4 folds, validate on 1, rotate — average the 5 scores

The confusion matrix

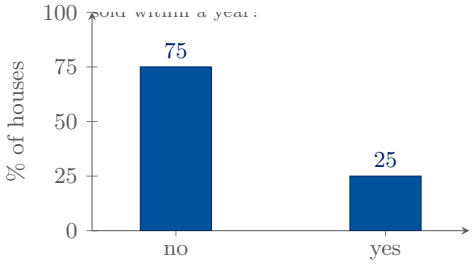
	Pred. no	Pred. yes
Actual no	750	0
Actual yes	250	0

"always predict no" ⇒ accuracy 75%, recall 0%

TP said yes, right
FP said yes, wrong
FN said no, wrong
TN said no, right

Accuracy, and why it lies

- regression
- classification



$$\text{accuracy} = \frac{TP + TN}{TP + FP + FN + TN}$$

say "no" every time: $750/1000 = 75\%$ right —

and you found **0** sales.

CATALYST

Precision and recall

precision $\frac{TP}{TP + FP}$

recall $\frac{TP}{TP + FN}$

precision: of the houses I said would sell, how many did?

recall: of the houses that sold, how many did I catch?

F1 and AUC

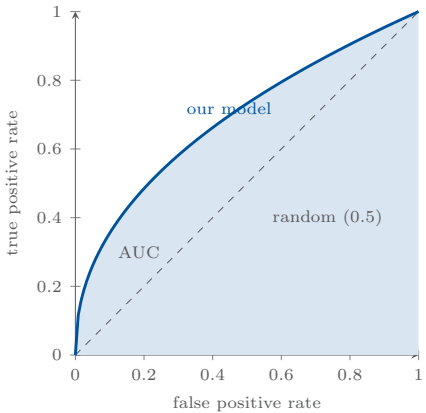
- regression
- classification

$$F_1 = \frac{2PR}{P + R}$$

one number that balances the two

AUC

ranks sold above not-sold, wherever you cut



CATALYST



Thank you

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